JINSEONG PARK

jinseong@kias.re.kr | https://jinseongp.github.io/ | jinseongP | scholar

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Summary

Affiliation	Korea Institute for Advanced Study (KIAS), Seoul, Republic of Korea
	Center for AI and Natural Sciences, AI Research Fellow (Mar 2025 - Present)
Research Topics	Trustworthy AI and Industrial Applications

Research Positions

Mar 2025 - Present	AI Research Fellow, Korea Institute for Advanced Study (KIAS) - Center for AI and Natural Sciences (CAINS), Seoul, Republic of Korea - Supervisor: Jeong Han Kim
Jun 2025 - Jul 2025	Visiting Postdoctoral Fellow, University of British Columbia (UBC) - Computer Science, Vancouver, British Columbia, Canada - Host: Mi Jung Park

Field of Interest

Safe and Trustworthy AI

- Privacy-Preserving Machine Learning
  - Generalization with Differential Privacy: C6, C3
  - Differential Private Machine Learning: C9, J2, P9
- Trustworthy of Generative Models
  - Machine Unlearning in Diffusion Models: P11, P10
  - Privacy Leakage in LLMs: C8
  - Social Impact of Diffusion Models: C6, C5, P4
- Generalization and Safety in Deep Learning
  - Generalization with Loss Landscape Analysis: C4, C3, C2, C1, P2, P1
  - Adversarial Attacks and Defenses: C4, C1, J3, J1

Industrial AI

- Reliable Time Series Analysis
  - Synthetic Data Generation for Time Series Data: P8
  - Robust Time Series Forecasting: C7, J4
- Industrial Domain Applications
  - Financial Applications: J5, W1, P7
  - Manufacturing Applications: Color cosmetics project (COSMAX), P6, P5
  - Explainability: P3

Note. Publications with overlap. J: journal, C: conference, W: workshop, and P: preprint.

Education

Mar 2022 - Feb 2025 Ph.D. in Industrial Engineering, Seoul National University
- [🎓Thesis] Generalization Frameworks for Temporal and Private Deep Learning: From Synthetic Data Generation to Robust Optimization
- Supervisor: Jaewook Lee
- Cumulative Major GPA: 3.94/4.30

Mar 2020 - Feb 2022 Master of Science in Industrial Engineering, Seoul National University
- [Thesis] Differentially Private Multi-class Classification Using Kernel Supports and Equilibrium Points
- Supervisor: Jaewook Lee
- Cumulative Major GPA: 4.19/4.30

Mar 2019 - Aug 2019 Student Exchange Program, Technische Universität Graz (TU Graz)
- Bachelor's program in Mechanical Engineering, Graz, Austria

Mar 2016 - Feb 2020 Bachelor of Science in Industrial and Management Engineering, Pohang University of Science and Technology (POSTECH)
- Graduated with Honors: Summa Cum Laude (rank: 1/22)
- Cumulative GPA: 3.91/4.30, Cumulative Major GPA: 4.03/4.30

Publications

†: equal contribution. *: corresponding author.

J: JOURNALS

[5] **Jinseong Park**, Hyungjin Ko, Jaewook Lee, "Modeling Asset Price Process: An Approach for Imaging Price Chart with Generative Diffusion Models". In: *Computational Economics* (2024), pp. 1–27.

[4] **Jinseong Park**, Hoki Kim, Yujin Choi, Woojin Lee, Jaewook Lee, "Fast sharpness-aware training for periodic time series classification and forecasting". In: *Applied Soft Computing* (2023), p. 110467.

[3] Hoki Kim, **Jinseong Park**, Jaewook Lee, "Generating Transferable Adversarial Examples for Speech Classification". In: *Pattern Recognition* 137 (2023), p. 109286.

[2] **Jinseong Park**, Yujin Choi, Junyoung Byun, Jaewook Lee, Saerom Park, "Efficient differentially private kernel support vector classifier for multi-class classification". In: *Information Sciences* 619 (2023), pp. 889–907.

[1] Yujin Choi, **Jinseong Park**, Jaewook Lee, Hoki Kim, "Exploring Diverse Feature Extractions for Adversarial Audio Detection". In: *IEEE Access* 11 (2023), pp. 2351–2360.

C: CONFERENCES

[9] **Jinseong Park**, Yujin Choi, Jaewook Lee, "Multi-Class Support Vector Machine with Differential Privacy". In: *Advances in Neural Information Processing Systems* (2025).

[8] Yujin Choi, Youngjoo Park, Junyoung Byun, Jaewook Lee, **Jinseong Park***, "Safeguarding Privacy of Retrieval Data against Membership Inference Attacks: Is This Query Too Close to Home?" In: *Conference on Empirical Methods in Natural Language Processing (EMNLP) Findings* (2025).

[7] Dongbin Kim, **Jinseong Park**, Jaewook Lee, Hoki Kim, "Are Self-Attentions Effective for Time Series Forecasting?" In: *Advances in Neural Information Processing Systems* (2024).

[6] **Jinseong Park**†, Yujin Choi†, Jaewook Lee, "In-distribution Public Data Synthesis with Diffusion Models for Differentially Private Image Classification". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2024).

[5] Yujin Choi†, **Jinseong Park**†, Hoki Kim, Jaewook Lee, Saerom Park, "Fair Sampling in Diffusion Models through Switching Mechanism". In: *Proceedings of the AAAI Conference on Artificial Intelligence* (2024).

[4] Hoki Kim, **Jinseong Park**, Yujin Choi, Jaewook Lee, "Fantastic Robustness Measures: The Secrets of Robust Generalization". In: *Advances in Neural Information Processing Systems* (2023).

[3] **Jinseong Park**, Hoki Kim, Yujin Choi, Jaewook Lee, "Differentially Private Sharpness-Aware Training". In: *International Conference on Machine Learning* (2023).

[2] Sungyoon Lee, **Jinseong Park**, Jaewook Lee, "Implicit Jacobian regularization weighted with impurity of probability output". In: *International Conference on Machine Learning* (2023).

[1] Sungyoon Lee, Woojin Lee, **Jinseong Park**, Jaewook Lee, "Towards Better Understanding of Training Certifiably Robust Models against Adversarial Examples". In: *Advances in Neural Information Processing Systems* 34 (2021), pp. 953–964.

W: WORKSHOPS

[1] Seungju Lee, **Jinseong Park**, Jaewook Lee, "Enhancing non-linear asset volatility forecasting models with investor sentiment and explainable AI". In: *Korea Computer Congress (KCC) 2023, XAI workshop, Best paper awards (2/37)* (2023).

P: PREPRINTS

[11] **Jinseong Park**, Mijung Park, "Data Unlearning Beyond Uniform Forgetting via Diffusion Time and Frequency Selection". In: (2025). Under review, ICLR 2026.

[10] Miso Kim, Georu Lee, Yunji Kim, Hoki Kim*, **Jinseong Park***, Woojin Lee*, "Co-occurring Associated REtained concepts in Diffusion Unlearning". In: (2025). Under review, ICLR 2026.

[9] **Jinseong Park**, Yujin Choi, Youngjoo Park, Jaewook Lee, "Privacy-Preserving Asymmetric Banach Kernel Encoding for Support Vector Classification". In: (2025). Under review, Pattern Recognition.

[8] **Jinseong Park**†, Seungyun Lee†, Woojin Jeong, Yujin Choi, Jaewook Lee, "Leveraging Priors via Diffusion Bridge for Time Series Generation". In: *arXiv preprint arXiv:2408.06672* (2024). Resubmission, KDD 2026.

[7] **Jinseong Park**, Jaewook Lee, Hyungjin Ko, "Advanced Market-Adaptive Portfolio Insurance: Dynamic Regime Detection Approach". In: *Available at SSRN 4779183* (2024). Resubmission, Economic Modelling.

[6] Yujin Choi, **Jinseong Park**, Youngjoo Park, Jaewook Lee, "Multiple Hyperplanes with Dual Encoding Support Vector Machine for Industrial Anomaly Detection". In: (2025). Under review, IEEE Transactions on Industrial Informatics.

[5] Yujin Choi, **Jinseong Park**, Youngjoo Park, Junyoung Byun, "Kernel-Based Differentially Private Upsampling for Imbalanced Datasets". In: *Available at SSRN 5118127* (2025). Under review, Engineering Applications of Artificial Intelligence.

[4] Yujin Choi, **Jinseong Park**, Junyoung Byun, Jaewook Lee, "Leveraging Programmatically Generated Synthetic Data for Differentially Private Diffusion Training". In: *arXiv preprint arXiv:2412.09842* (2024). Under review, AAAI 2026.

[3] Hoki Kim, **Jinseong Park**, Yujin Choi, Seungyun Lee, Jaewook Lee, "BayesNAM: Leveraging Inconsistency for Reliable Explanations". In: *arXiv preprint arXiv:2411.06367* (2024). Under review, Pattern Recognition.

[2] Hoki Kim, **Jinseong Park**, Yujin Choi, Woojin Lee, Jaewook Lee, "Exploring the effect of multi-step ascent in sharpness-aware minimization". In: *arXiv preprint arXiv:2302.10181* (2023).

[1] Hoki Kim, **Jinseong Park**, Yujin Choi, Jaewook Lee, "Stability analysis of sharpness-aware minimization". In: *arXiv preprint arXiv:2301.06308* (2023). Under review, IEEE Transactions on Pattern Analysis and Machine Intelligence.

Patents

[Patent Application] 10-2023-0087024, Korea Jun 2023

- 적대적 음성 예제를 감지 방법 및 시스템 (METHOD AND SYSTEM FOR DETECTING ADVERSARIAL SPEECH EXAMPLE)

[Patent Application] 10-2023-0087006, Korea Jul 2023

- 적대적 음성 예제 생성 방법 및 시스템 (METHOD AND SYSTEM FOR GENERATING ADVERSARIAL SPEECH EXAMPLE)

[Patent Application] 10-2023-0073178, Korea Jul 2023

- 커널 서포트 벡터에 기반한 차분 프라이버시 다중 분류 학습을 위한 장치 및 방법 (Apparatus and Method for Learning Differential Privacy Multi-Classification Based on Kernel Support Vector)

Invited Talks

Trustworthy Generative AI

- KIAS Center for AI and Natural Sciences Workshop, Jeongseon, Korea May 30, 2025
- Korean Society for Industrial and Applied Mathematics (KSIAM), Seoul, Korea May 17, 2025

Privacy-Preserving Deep Learning

- UBC Computer Science Park Lab ML Seminar, Vancouver, Canada Jul 23, 2025
- UNIST Department of Industrial Engineering, Ulsan, Korea Jun 11, 2025
- KIAS School of Computational Sciences Workshop, Gangneung, Korea May 22, 2025
- KIAS Center for AI and Natural Sciences Seminar, Seoul, Korea Mar 23, 2025
- Desilo, Seoul, Korea Jul 2024

Academic Experiences

Academic Projects, (P): as a PI

- (P) Development of Generative Models for Safe and Trustworthy AI[1], Korea Institute of Advanced Study (KIAS), Ministry of Science and ICT — Mar, 2025 - Present

Seminar Hosts

- Hoki Kim (Chung-Ang University), KIAS CAINS Seminar — Sep 24, 2025
- Jongheon Jeong (Korea University), KIAS CAINS Seminar — Sep 11, 2025
- Lai, Chieh-Hsin (Sony AI), KIAS CAINS Seminar — Sep 03, 2025

Academic Reviewers

- Journal: Pattern recognition, IEEE Internet of Things
- Conference: CVPR, NeurIPS, ICML, AAAI, KDD

Attendee, Algorithmic Fairness in Graduate Summer School, LA, USA — Jul 2022

- Hosted by Institute for Pure & Applied Mathematics (IPAM) at UCLA
- Organized by Cynthia Dwork, Guy Rothblum, and Noa Dagan

Lab Research Intern, IDEA Lab, Pohang, Korea — Sep 2019 - Dec 2019

- Industrial Data Engineering & Analytics Lab, hosted by Prof. Hyunbo Cho at POSTECH

Teaching Experiences

Lecturer, Soongsil University — Fall 2024

- Analysis of Financial Time Series (금융시계열분석) (2024-2, 2150052601, 3 credits), Seoul, Korea
- Teaching evaluation score: 96.4/100.0 (average: 88.3/100.0)

Teaching Assistant (TA), Industrial Engineering (Prof. Jaewook Lee), SNU — 2020 - 2023

- Undergraduate courses: Statistics for Industrial Engineering (Spring 2020, Spring 2022, and Spring 2023), Mathematical Methods for Industrial and Management Engineering (Fall 2020)
- Graduate courses: Data Mining Technology (Fall 2022), Advanced Topics in Statistical Learning (Spring 2023)

TA & Instructor, Corporate Education Programs, SNU — 2020 - 2023

- Instructor for big data analysis with Python and deep learning with PyTorch in Woori Bank (2022)
- Instructor for AI practice course in Korea Institute of Startup & Entrepreneurship Development (KISED) (2022)
- TA for Machine Learning and Deep Learning for Prof. Jaewook Lee in SNU Big-Data FinTech course (2021 and 2023)
- TA for Python programming for Prof. Namhyung Kim in SNU Big-Data FinTech course (2022)
- TA for Optimization and Linear Algebra for Prof. Jaewook Lee in Samsung Data Scientist for Device Solution (2020-2022) and Industrial Bank of Korea (2020 and 2021)

[1] 안전하고 신뢰가능한 인공지능을 위한 생성 모델 개발

Industrial Experiences

[Project] COSMAX, Seoul, Korea Dec 2023 - Feb 2024, Jun 2024 - Sep 2024

- Lead Researcher, Prediction and Generative Neural Networks for Color Cosmetics

[Intern] SK Telecom, Seoul, Korea Jun 2018 - Aug 2018

- R&D Digital Transformation Team Intern
- Developed services for R&D Digital Transformation of internal data clusters

[Project] LNG industry, POSCO Energy, Pohang, Korea 2018

- Student Researcher, with Prof. Ribin Seo at POSTECH
- Organized LNG value chains and strategic implications of POSCO Energy

Scholarship

2023	The 15th Lee Joong Han Award (Research), Industrial Engineering, SNU.
2022 - 2023	Teaching Assistant scholarship, SNU, Full tuition for all semesters.
2021 Fall	Scholarship for Academic Excellence, SNU, Merit-based, 30% of tuition for a semester.
2019 Spring	Scholarship for Exchange Students, POSTECH.
2018 Fall, 2019 Fall	Mentor Scholarship for IME Information System Tech, C++ programming, POSTECH.
2018 - 2019	National Scholarship for Science and Engineering, Republic of Korea, Merit-based, full tuition for all semesters.
2016 - 2017	Scholarship for Academic Excellence, POSTECH, Merit-based, full tuition for all semesters.

References

Jaewook Lee Professor of Industrial Engineering, Seoul National University
1 Gwanak-ro, Gwanak-gu, Seoul, KR
jaewook@snu.ac.kr | https://safeai.snu.ac.kr

Saerom Park Assistant professor of Industrial Engineering, UNIST
50, UNIST-gil, Eonyang-eup, Ulju-gun, Ulsan, KR
srompark@unist.ac.kr | https://sites.google.com/view/safe-ai-lab

Sungyoon Lee Assistant professor of Computer Science, Hanyang University
Room 403-2, IT/BT Building, Wangsimni-ro 222, Seongdong-gu, Seoul, KR
sungyoonlee@hanyang.ac.kr | https://sites.google.com/view/sungyoon-lee